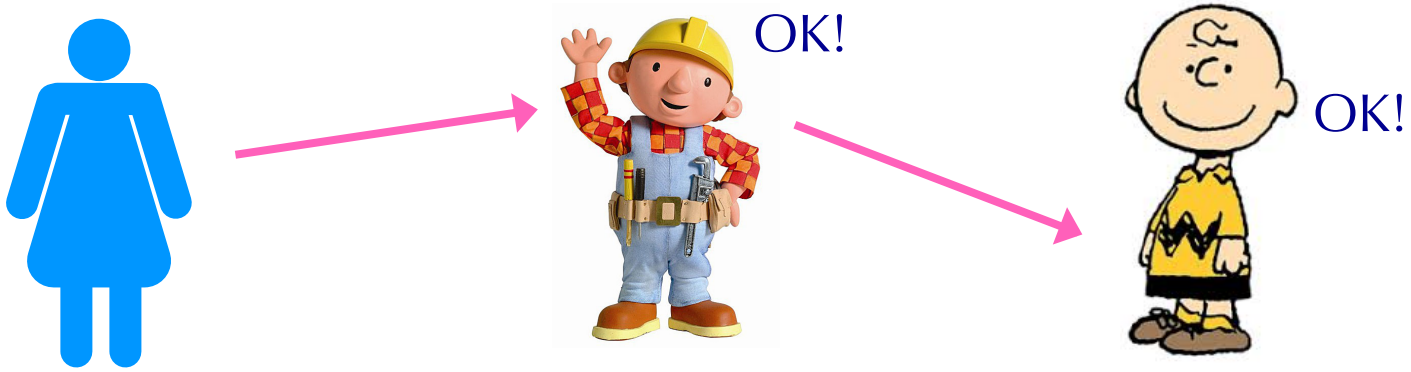
Unconditionally secure quantum signatures



Theory:
Petros Wallden, Ryan Amiri, Ittoop Puthoor, Juan Miguel Arrazola, Jonathan Crickmore, Vedran Dunjko, John Jeffers, Erika Andersson
Experiments:
Robert Collins, Ross Donaldson, Klaudia Kleczkowska, Patrick Clarke, Gerald Buller

What are signature schemes?

Widely used! Different from encryption.



Alice (sender) can send a signed message, so that Bob (recipient) can be sure that the message is from Alice.

Message is transferrable:
Bob can know that if he forwards the message to Charlie, then Charlie will also accept the message as genuinely from Alice.

No forging: Messages cannot be forged.
Non-repudiation: Alice cannot deny having sent a message.

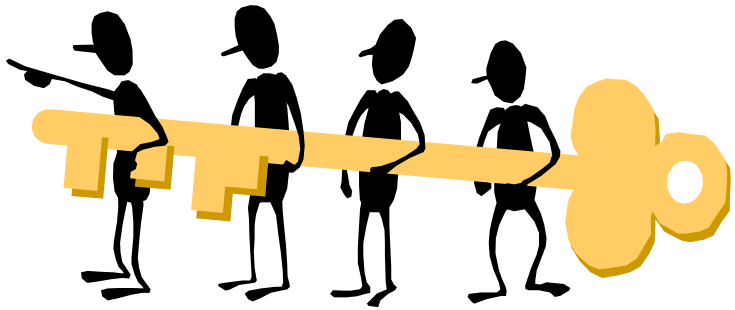
"Classical" public-key digital signatures: computational security

Can use one-way functions:

$x \rightarrow f(x)$ easy – encryption

$f(x) \rightarrow x$ hard – decryption

$f(x)$ publicly known: "public key".



Public key crypto relies on the fact that factoring large numbers or calculating discrete logarithms is "hard".

No "classical" function has been proven to be hard to invert.

Signatures with unconditional or information-theoretic security

"Classical" schemes exist, but need secret channels AND either

- an authenticated broadcast channel

  Chaum and Roijakkers (Crypto '90, LNCS, **537**, 206 (1991))

- a trusted initializer to distribute signing keys and verification keys to participants.

  Hanaoka et al. (Asiacrypt 2000, LNCS, **1976**, 130 (2000))



Useful paper: Swanson & Stinson,
Unconditionally secure signature schemes revisited,
ICITS 2011, LNCS, **6673**, 100 (2011).

Unconditional quantum security

The first quantum signature schemes resembled public-key signatures (starting with Gottesman & Chuang, arXiv:0105032)…

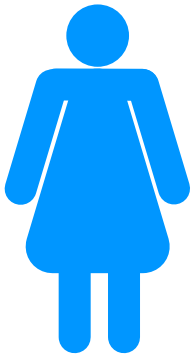
…these have evolved into schemes using QKD equipment, or even using standard QKD (Wallden et al., PRA **91**, 042304 (2015); Amiri et al., arXiv:1507.02975 (2015).



Quantum signatures

Basic principle (many variations possible)
Two non-orthogonal quantum states $|a\rangle$ and $|b\rangle$.

Alice $|a\rangle|b\rangle|b\rangle|a\rangle|b\rangle|a\rangle...$





Bob measures:
a, fail, b, fail, b, a…

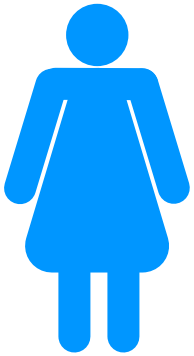
$|a\rangle|b\rangle|b\rangle|a\rangle|b\rangle|a\rangle...$



Charlie measures:
a, b, b, fail, fail, a…

Alice has *full information*.
Recipients Bob and Charlie have *partial information*.
To sign: Present *a, b, b, a, b, a, a…*

Quantum signatures (more details)



Distribution stage to set up:
Alice wants to later sign a 1-bit message.
Her "quantum signatures" are randomly selected sequences of quantum states

$$|\psi^0_{i1}\rangle \otimes |\psi^0_{i2}\rangle \otimes \cdots \otimes |\psi^0_{iM}\rangle \qquad \text{for "0"}$$

$$|\psi^1_{i1}\rangle \otimes |\psi^1_{i2}\rangle \otimes \cdots \otimes |\psi^1_{iM}\rangle \qquad \text{for "1"}$$

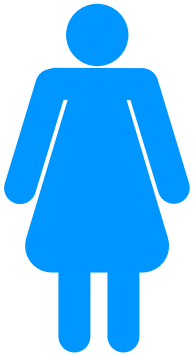


which she sends to Bob and Charlie.
Nobody except Alice knows exactly what these states are.



Gottesman & Chuang, arXiv:0105032
EA, M. Curty & I. Jex, PRA **74**, 022304 (2006)
Clarke et al., Nature Communications 3, 1174 (6 Nov 2012)
Dunjko et al., PRL **112**, 040502 (2014)
Collins et al., PRL **113**, 040502 (2014)
Yin et al., arXiv:1507.03333

Quantum secure signatures



Alice's "quantum signatures" are

$$|\psi^0_{i1}\rangle \otimes |\psi^0_{i2}\rangle \otimes \cdots \otimes |\psi^0_{iM}\rangle \qquad \text{for "0"}$$

$$|\psi^1_{i1}\rangle \otimes |\psi^1_{i2}\rangle \otimes \cdots \otimes |\psi^1_{iM}\rangle \qquad \text{for "1"}$$



Messaging stage (later, entirely classical):
Present message together with classical info
about corresponding sequence of states.
Only Alice knows!
Bob and Charlie check for mismatches.



Gottesman & Chuang, arXiv:0105032
EA, M. Curty & I. Jex, PRA **74**, 022304 (2006)
Clarke et al., Nature Communications 3, 1174 (6 Nov 2012)
Dunjko et al., PRL **112**, 040502 (2014)
Collins et al., PRL **113**, 040502 (2014)
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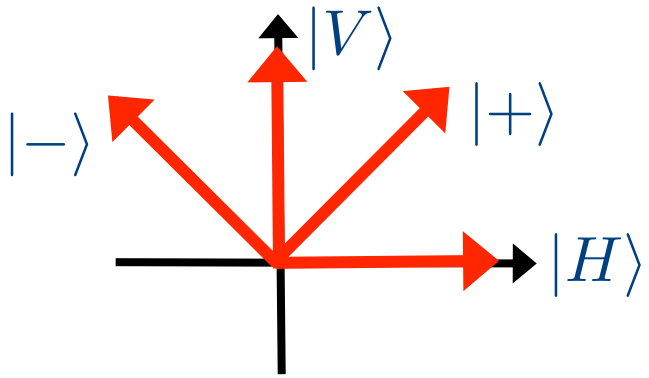
Quantum "one-way" function

Set of non-orthogonal quantum states $|\psi_k\rangle$

$k \rightarrow |\psi_k\rangle$ easy

$|\psi_k\rangle \rightarrow k$ hard

e.g. BB84 states



Provably "hard"!
Information about *k* is limited by the Holevo bound:

$$I_{acc} \leq \chi(\rho_{key}) = S(\rho_{key}) - \sum_k p_k S(\rho_k)).$$

$$\rho_{key} = \sum_k p_k \rho_k, \quad S(\rho_{key}) = -\mathrm{Tr}(\rho_{key} \log \rho_{key}).$$

Quantum "one-way" function

Set of non-orthogonal quantum states $|\psi_k\rangle$

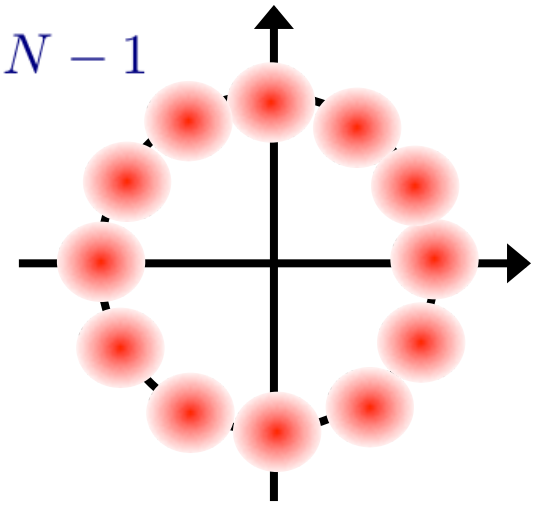
$k \rightarrow |\psi_k\rangle$ easy

$|\psi_k\rangle \rightarrow k$ hard

or coherent states $|\alpha_k\rangle$

$\alpha_k = |\alpha| e^{ik2\pi/N}$

$k = 0, 1, ..., N-1$



Provably "hard"!

Information about *k* is limited by the Holevo bound:

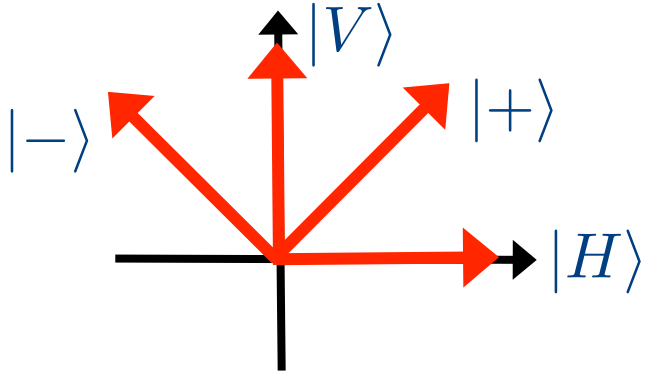
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$$\rho_{key} = \sum_k p_k \rho_k, \quad S(\rho_{key}) = -\mathrm{Tr}(\rho_{key} \log \rho_{key}).$$

No need for quantum memory!

Bob and Charlie immediately measure states, obtaining partial information.

E.g. for BB84 states,
can rule out 1 of 4 states
with 100% certainty.



To send a signed message, a forger must avoid declaring states that have been ruled out.

Probability to forge $\sim e^{-const \cdot L}$, $L =$ ***signature length***

Dunjko et al., PRL **112**, 040502 (2014) **Open access**
Collins et al., PRL **113**, 040502 (2014) **Open access**

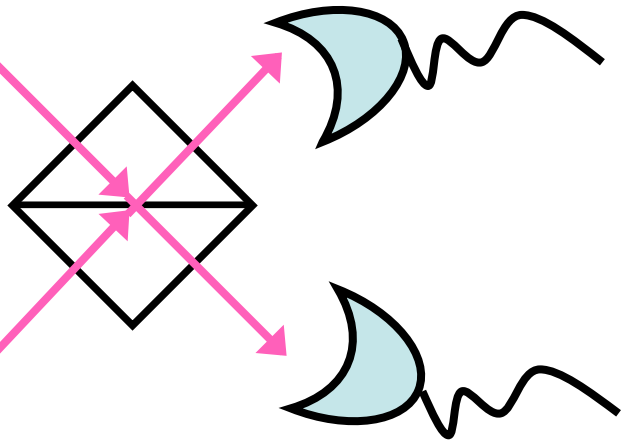
Quantum state elimination for coherent states

Known reference state $|\alpha\rangle_1$, unknown state $|\beta\rangle_2$

In:

$|\alpha\rangle_1$

$|\beta\rangle_2$



Out:

$|(\alpha+\beta)/\sqrt{2}\rangle_1$

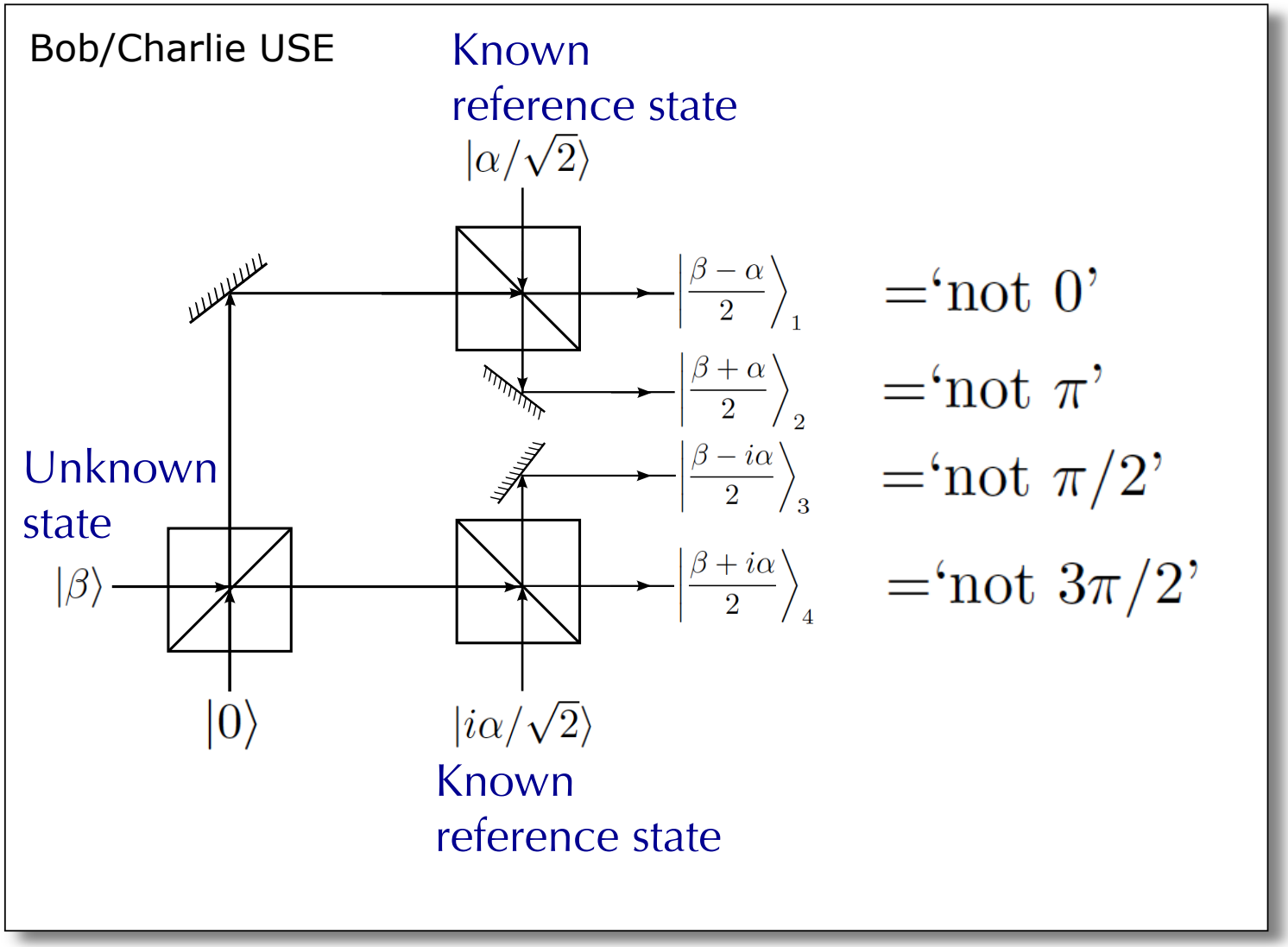
$|(\alpha-\beta)/\sqrt{2}\rangle_2$

If we see any light in output mode 2, then $\alpha \neq \beta$.

If we see any light in output mode 1, then $\alpha \neq -\beta$.

State elimination with 4 possible coherent states

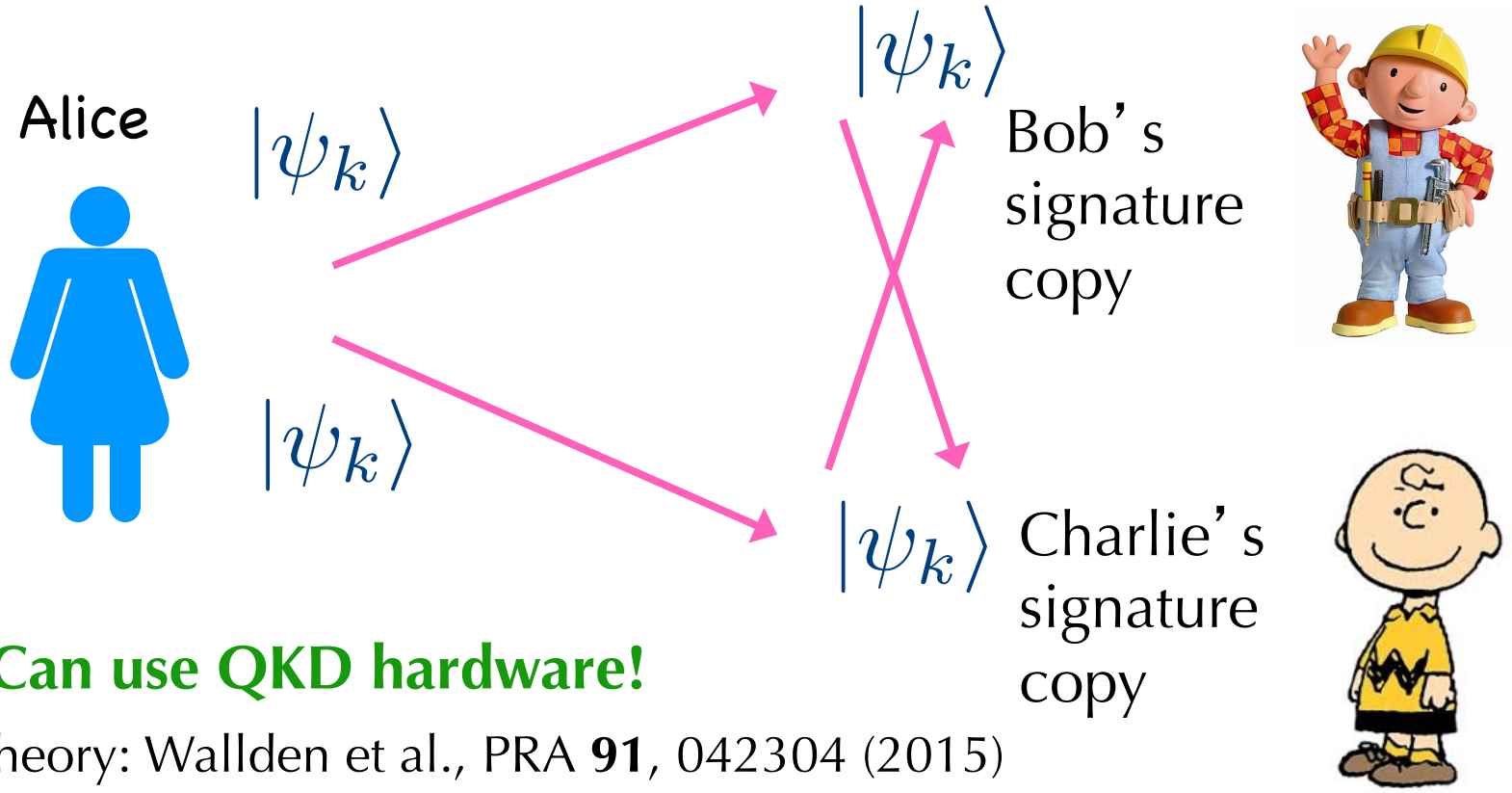
$$|\alpha\rangle, |i\alpha\rangle, |-\alpha\rangle, |-i\alpha\rangle$$



Making sure Alice does not cheat

Bob randomly selects half of the signature states, and forwards either states or measurement results to Charlie.

Must be done in secret from Alice!



Can use QKD hardware!

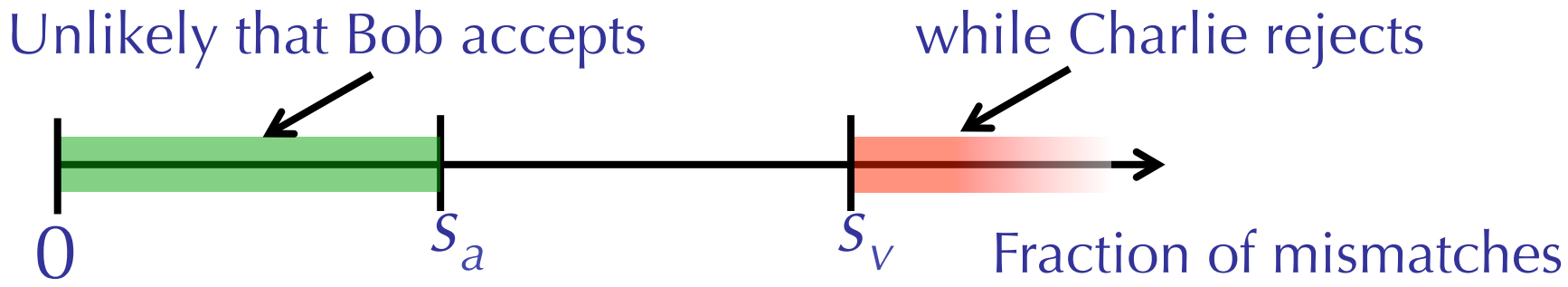
Theory: Wallden et al., PRA **91**, 042304 (2015)

Expt: Donaldson et al., arXiv:1509.07827 (2015)

Making sure Alice does not cheat

Bob and Charlie symmetrize their measurement records.
Either both accept or both reject a message.

Important: $s_a < s_v$: Must have this for USS schemes in general!
s_a = threshold for accepting a message received from Alice
s_v = threshold for accepting a forwarded message

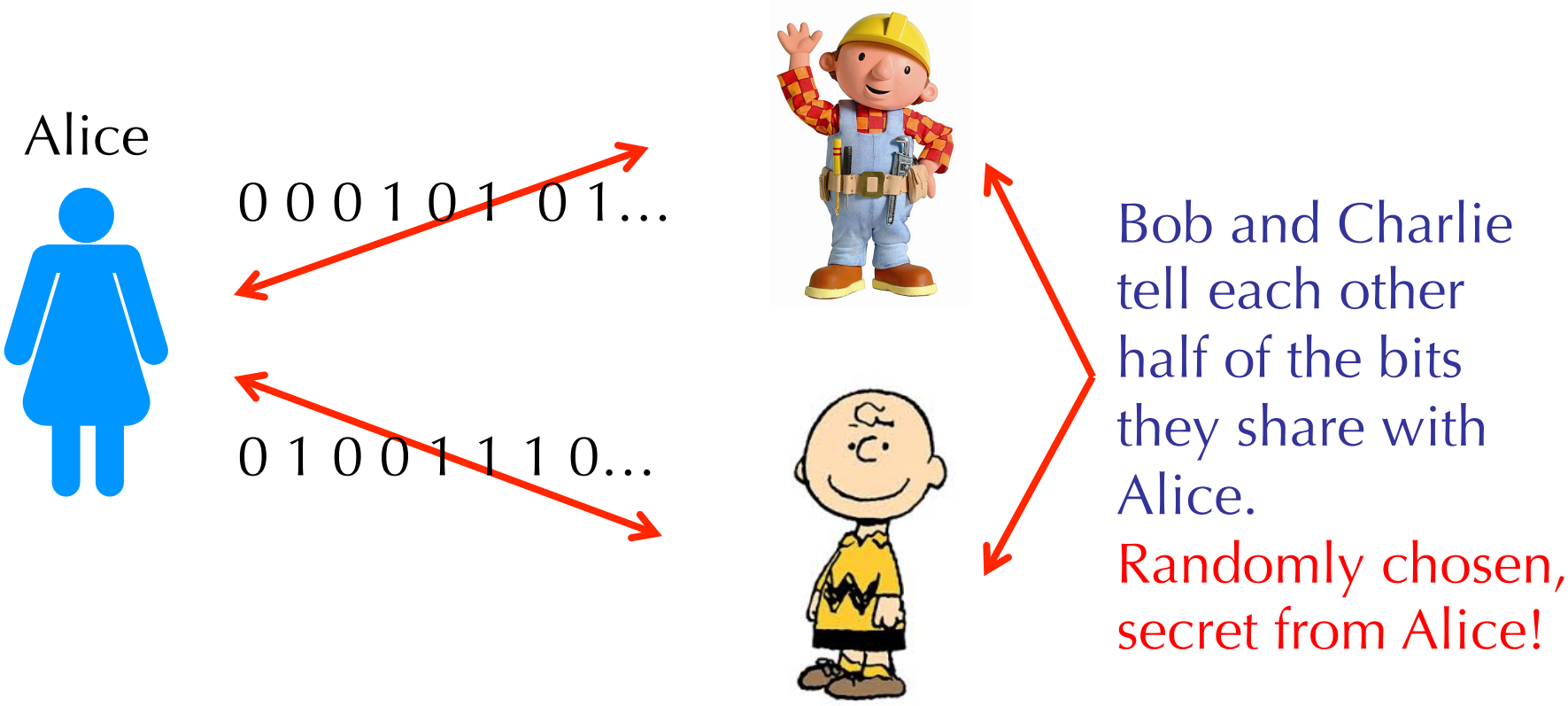


Probability to cheat $\sim e^{-const \cdot L}, L =$ ***signature length***

“Classical” signatures with unconditional or information-theoretic security

“P2” by Wallden et al., Phys. Rev. A **91**, 042304 (2015):

Setup phase: Alice shares a secret key with Bob, and a different secret key with Charlie.



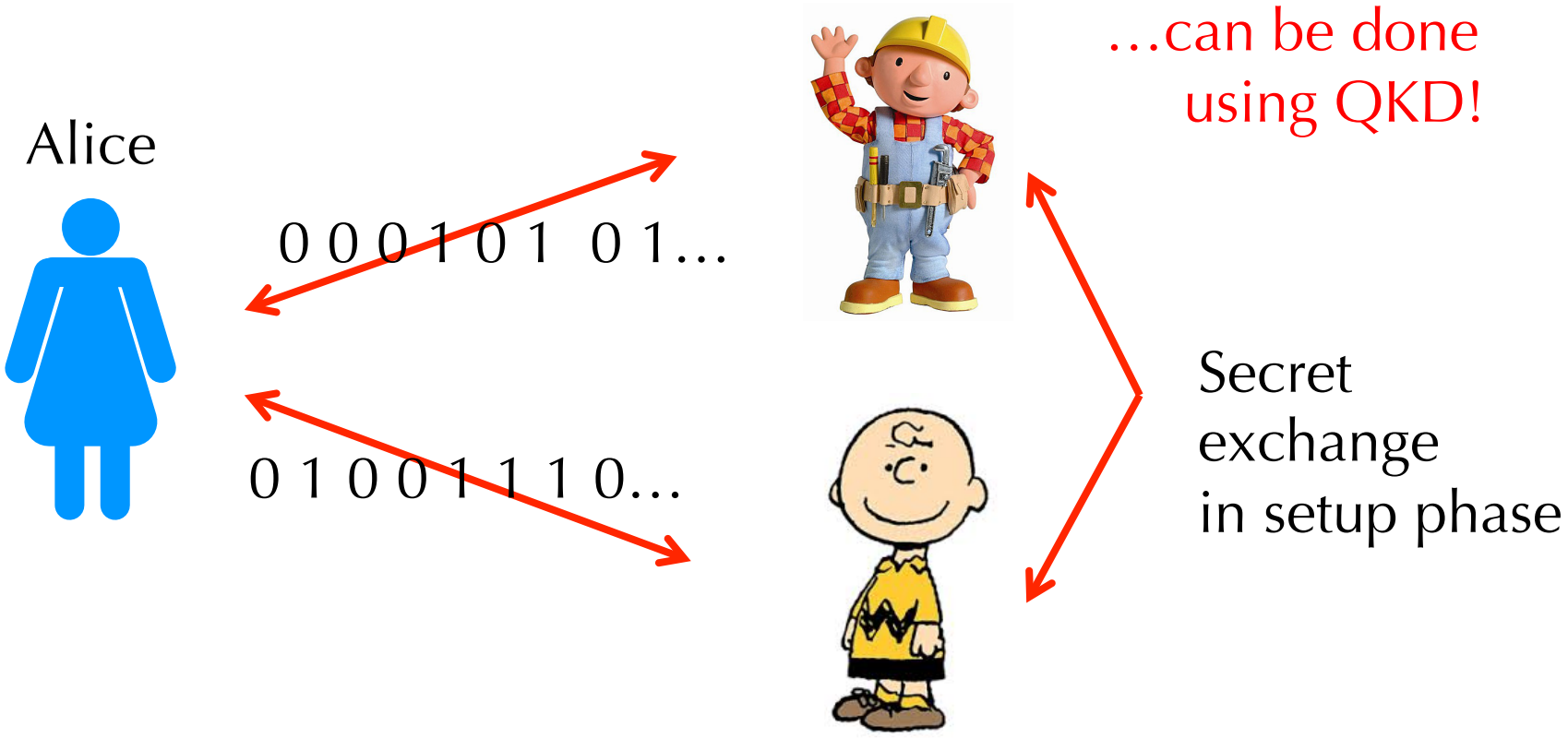
“Classical” signatures with unconditional or information-theoretic security

“P2” by Wallden et al., Phys. Rev. A **91**, 042304 (2015):

To sign: Present message and all its corresponding key bits.

Resources: Pairwise secret communication channels….

…can be done using QKD!



Quantum secure signatures and QKD

Why not just do usual QKD plus classical signature scheme?



Efficiency:

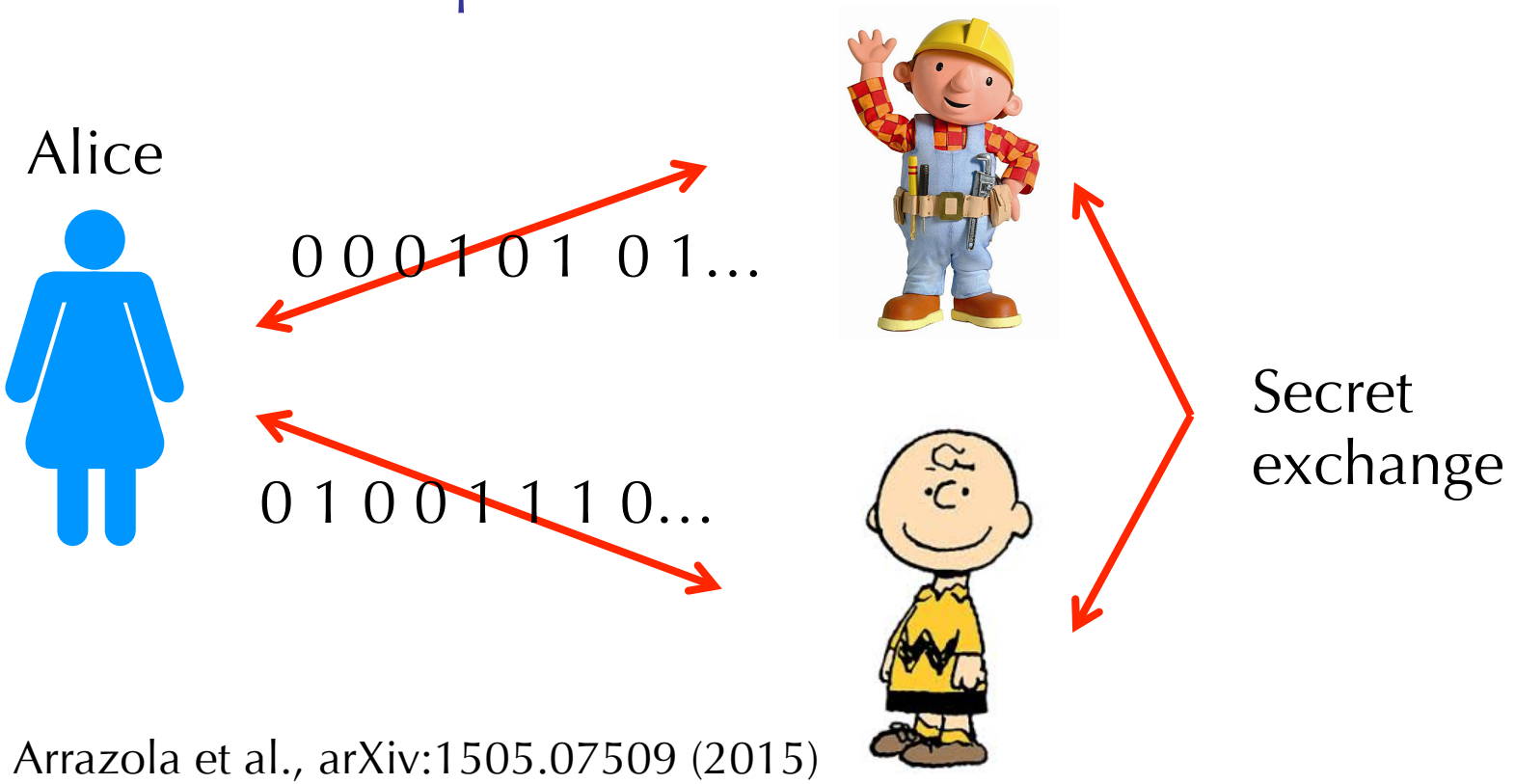
- QBER threshold different in practice for QKD and signatures.
  Do parameter estimation, but can stop before error correction and privacy amplification.
  Amiri et al., arXiv:1507.02975 (2015)

- # of quantum channels needed can be lower for "direct" quantum signatures.

Some necessary properties of unconditionally secure signature schemes

-Cannot be deterministic! Need shared secrets.

For example:



Arrazola et al., arXiv:1505.07509 (2015)

Some necessary properties of unconditionally secure signature schemes

-No limit on computations by users:
Users *must* have different verification procedures.
(e.g. s_a, s_v, or if more recipients, s_1, s_2, s_3, …,
checking different parts of Alice's signature…)





Arrazola et al., arXiv:1505.07509 (2015)

Some necessary properties of unconditionally secure signature schemes

-Need to think about dispute resolution.
(e.g. trusted arbiter, majority vote, weighted vote)

-Need to think about trust assumptions
(e.g. at least half of participants honest for majority vote)



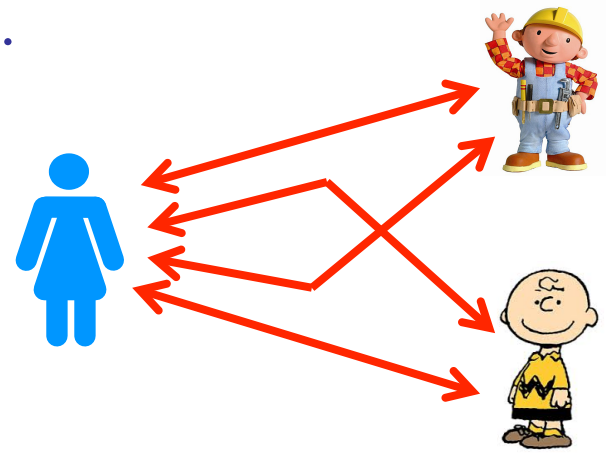
Arrazola et al., arXiv:1505.07509 (2015)

What leads to what?

Can do signatures, if either

-*Untraceable sending:* Participants can send Alice information such that she does not know who sent what.

-*Oblivious transfer (sort of):* Alice can send participants information in such a way that she does not know who ends up with what.



Quantum secure signatures – team members

Theory:
Petros Wallden, Ryan Amiri, Ittoop Puthoor, Juan Miguel Arrazola, Jonathan Crickmore, Vedran Dunjko, John Jeffers, EA
Experiments:
Robert Collins, Ross Donaldson, Klaudia Kleczkowska, Patrick Clarke, Gerald Buller

Clarke et al., Nature Communications 3, 1174 (6 Nov 2012) **Open access**
Dunjko et al., PRL **112**, 040502 (2014) **Open access**
Collins et al., PRL **113**, 040502 (2014) **Open access**
Wallden et al., PRA **91**, 042304 (2015) **Open access**
Review paper: Amiri and Andersson, Entropy **17**, 5635 (2015) **Open access**
Arrazola et al., arXiv:1505.07509 (2015)
Amiri et al., arXiv:1507.02975 (2015)
Donaldson et al., arXiv:1509.07827 (2015)











